Learning 3D Robotic Manipulation from Demonstrations: A Controlled Comparison of Behavior Cloning and Diffusion Policies

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RoboForge technical report (local build, not peer reviewed)

Abstract

We ask whether a conditional diffusion policy over action chunks controls a simulated articulated robot more reliably than deterministic and recurrent behavior cloning when all of them learn from the same demonstrations. On the robosuite Lift and PickPlaceCan tasks (MuJoCo physics, a Panda arm with operational-space control) we train four policy families on the robomimic proficient-human demonstrations with exact simulator state as input: a one-step MLP, an LSTM, an MLP that predicts action chunks, and a 1D temporal U-Net diffusion policy, with matched parameter counts (1,077,255 to 1,204,336), 3 training seeds and a matched, preregistered tuning budget. Checkpoints are chosen on validation rollouts, and the frozen protocol is then run once on 50 held-out initial states per task. On Can, diffusion succeeds in 95.3 % of held-out episodes against 76.7 % for the one-step MLP (paired difference +18.7 pp, 95 % interval [+4.7, +32.7]), although a seed-level t interval, computed after the audit and not preregistered, includes zero; on Lift the same difference is not separated by the preregistered bootstrap (the seed-level interval does separate it). Diffusion is not separated from the recurrent policy or from the chunked MLP on either task, so these data give no evidence that the generative model adds to observation history and action chunking, while diffusion costs 19–26× the chunked MLP's training time. Under preregistered perturbations of the initial arm configuration, diffusion degrades least on Lift, but on Can its drop is not separated from the chunked MLP's (+4.7 pp [−10.7, +20.7]); the diffusion sampler's bound on its actions is a confound in these comparisons. Two separately frozen extensions follow: on Lift, visual diffusion and visual chunked BC from two cameras both reach the task's ceiling, and a DP3-inspired point-cloud diffusion policy generalizes worse than the frozen RGB diffusion policy (same head and training, different encoder and augmentation) to cube placements outside the demonstrations' range (−41.3 pp [−54.0, −28.0]). The main limitation is resolution: 3 seeds and 50 starts per task cannot resolve differences of a few points near the ceiling, and everything here is simulation.

1 Introduction

Learning robot control from demonstrations turns a manipulation skill into supervised learning: a policy is fit to the actions an operator took in the states the operator visited [2, 18]. Two difficulties make this harder than it looks. First, a policy that is slightly wrong drifts into states the demonstrations never covered, and its errors compound over a closed-loop episode [21]. Second, human demonstrations are often multimodal: several different action sequences solve the same situation, and a regressor trained with a squared loss averages them into an action that solves none [6, 22].

Diffusion policies [3] address both concerns directly. They model the conditional distribution of a whole chunk of future actions with a denoising diffusion model [9], so a multimodal distribution can be represented, and they execute the chunk in a receding-horizon loop, which reduces how often the policy must decide. Reported gains over behavior cloning (BC) are large on several benchmarks, but a gain of this kind can have more than one cause. A diffusion policy differs from a one-step BC policy in its generative training objective, in its observation history, in predicting action chunks rather than single actions, and often in its tuning budget. Action chunking alone is known to help deterministic policies [29].

This report describes a controlled comparison that separates these factors as far as a small budget allows. We compare four families on two robosuite tasks under one protocol: a one-step MLP (BC-MLP), an LSTM over an observation window (BC-RNN), an MLP with exactly the diffusion policy's observation and action horizons (BC-MLP-chunk), and a conditional 1D temporal U-Net diffusion policy. The chunked MLP is the control that the protocol requires before any advantage of diffusion over BC-MLP may be attributed to the generative model (hypothesis H2-c). All families see the same exact-state observations, use the same action encoding and parameter budget, and receive the same number of tuning trials; all are selected by the same rule on validation rollouts and tested once on held-out initial states after a committed freeze.

Contributions. The contribution is empirical and methodological, not a new algorithm:

- a preregistered, paired, closed-loop comparison of BC and diffusion policies on Lift and Can, with a chunked deterministic control, a held-out test run once, and every null result reported (Sections 6–7);
- preregistered robustness sets and ablations (exponential moving average, sampler, observation history), analysed with the same hierarchical paired bootstrap;
- two separately frozen observation-modality extensions on Lift: a visual BC versus visual diffusion comparison, and a DP3-inspired point-cloud diffusion policy against the frozen RGB diffusion policy under a cube-placement shift (Section 8);
- a reproducible pipeline: every number in this report is generated from hash-pinned frozen result files (Appendix A);
- an independent audit of the code and frozen files after the results were frozen, whose findings are disclosed with their consequences (Section 9), including a seed-level sensitivity check of every declared success comparison, an analysis that was not preregistered and is labelled as such.

We make no claim of a new diffusion method, of real-robot performance, or of superiority beyond what the preregistered paired intervals show.

2 Related work

Behavior cloning. Fitting a policy to demonstrated actions goes back to ALVINN [18]; surveys of learning from demonstration cover the wider field [2]. Compounding error under distribution shift motivates interactive methods such as DAgger [21]; here, as in most offline benchmarks, no interaction with an expert is available. Recurrent policies over observation histories, trained with LSTMs [10], are a strong baseline on the robomimic tasks [14]. Predicting chunks of future actions and executing them open loop for a few steps, as in ACT [29], reduces the effective decision horizon of a deterministic policy.

Multimodal and generative policies. Implicit (energy-based) BC [6] and Behavior Transformers [22] model multimodal action distributions without diffusion. Diffusion Policy [3] applies denoising diffusion [9] with DDIM sampling [23] to action chunks, using a 1D temporal U-Net in the style of Diffuser [11], itself derived from the U-Net [20], with FiLM conditioning [17]. Our diffusion family adapts that architecture and its noise schedule [16] from the robomimic reference implementation and the diffusers library [25].

Benchmarks and simulators. robomimic [14] provides the demonstrations and baseline recipes; robosuite [30] provides the tasks, robot and controllers on the MuJoCo physics engine [24]. We use the proficient-human datasets as repackaged for robosuite v1.5, which the dataset maintainers note may not reproduce the original benchmark numbers exactly; none of our numbers should be compared with the original paper's.

3D perception for policies. 3D Diffusion Policy (DP3) [28] conditions a diffusion policy on a compact encoding of a point cloud and reports better generalization than image-based policies. Its encoder descends from PointNet [19]. Our point-cloud arm is DP3-inspired: it follows the idea of a per-point MLP with max pooling, but it is not a reproduction of DP3's implementation, tasks or protocol.

Evaluation practice. Reports of learned control are sensitive to seeds and to how uncertainty is summarized [1, 8]. We resample both training seeds and initial states in a hierarchical bootstrap [4, 5], pair every comparison by initial state, and report per-checkpoint Wilson intervals [26].

3 Benchmark setup and data

Tasks and simulator. Both tasks run in robosuite 1.5.1 (commit 51cc017) on MuJoCo 3.2.7 with a Panda arm and robosuite's OSC_POSE operational-space controller at 20 Hz. In *Lift* the robot grasps a cube and raises it above a height threshold. In *Can* (robosuite PickPlaceCan) it picks a can from a source bin and places it in its target compartment of a second bin. Success is robosuite's task predicate, evaluated on the current kinematics after every control step (Section 9); an episode ends at the first success or after 400 control steps. Initial states are serialized MuJoCo snapshots (model XML, time, joint positions and velocities) that are restored exactly for every policy, so no evaluation depends on a random reset.

Demonstrations. We use the robomimic proficient-human (PH) low-dimensional datasets for both tasks, in the version repackaged for robosuite v1.5 (Hugging Face `robomimic/robomimic_datasets`, revision 74fa018; file SHA-256 prefixes 2067777c for Lift and 3f2eb92e for Can) [14]. Each task has 200 successful demonstrations: Lift episodes last 36–64 control steps (mean 48.3) and Can episodes 82–151 (mean 116.0). The data were recorded through one teleoperation channel, so no claim about held-out operators is possible.

Split. Demonstrations are split by whole episode with robomimic's upstream masks into 180 training and 20 validation episodes, with identical ID lists for both tasks. Normalization statistics are fit on the training episodes only. Training windows never cross an episode boundary, and no observation after the decision time is ever an input. The validation demonstrations give offline losses only: no model, configuration or checkpoint is chosen with them. There is no demonstration test split; the test is a set of held-out simulator initial states (Section 5).

Observations and privilege. Every Level 1 policy receives the same 19-dimensional vector per step: end-effector position and orientation (quaternion), the two gripper finger joint positions, the object's position and orientation, and the relative position of gripper and object. The object entries are exact simulator poses, so every Level 1 result is a *privileged-state* result and is never compared causally with camera-based policies. For Can we compute the relative position from the current step's poses; the dataset's stored relative-pose features lag by one step at the first frame, so our Can input is not robomimic's default feature set and our Can numbers are not comparable with numbers obtained on that set.

Actions. The action is the controller's 7-dimensional command: a position delta, an axis-angle orientation delta and a gripper command, each scaled to $[-1, 1]$ by the controller limits. The demonstrations' gripper commands are exactly ± 1. Every family predicts min–max normalized actions; predictions are mapped back with the inverse transform and clipped to the action bounds by one shared decoder.

Platform. Every reported rollout ran on one canonical evaluation platform: a Linux aarch64 container (image `sha256:`1f685638f28c, Python 3.11.18, OSMesa rendering) on an Apple M1 Pro host, with policy inference on the CPU with one thread so that rollouts are deterministic given their seeds. Physics is not bit-identical across platforms, so native macOS rollouts were used for development only and never reported.

4 Methods

4.1 Policy families

Table 1 lists the four Level 1 families. All share the observation of Section 3, the normalized action, a masked mean-squared error in normalized action space (the diffusion family's error is on the predicted noise), float32 training on the CPU and one shared action decoder.

- **BC-MLP** maps the current observation to the next action ($T_o = T_p = T_a = 1$), through hidden layers of width 1024.
- **BC-RNN** runs an LSTM [10] with hidden size 296 over the last 10 observations and predicts the next action from its last valid step. At every decision the LSTM restarts from a zero state over the window, exactly as in training, so no hidden state persists between decisions or episodes. This differs from robomimic's BC-RNN, which carries its hidden state for several steps; we chose exact train/inference consistency.
- **BC-MLP-chunk** keeps BC-MLP's hidden layers but takes the diffusion policy's window of $T_o = 2$ observations and predicts $T_p = 16$ future actions, of which it executes the first $T_a = 8$ before observing again. It shares the diffusion policy's inputs, horizons and training budget and is the control for H2-c. It differs in the head (a deterministic MLP against a denoising U-Net evaluated with EMA weights), in the action bound described below, and in the tuning trial each family selected.
- **Diffusion U-Net** is a conditional denoising diffusion model over the normalized $T_p \times 7$ action chunk [3]. The $T_o \times 19$ observation window is flattened into a global condition, with no learned observation encoder. A 1D temporal U-Net [11] with down widths (64, 128), GroupNorm [27] and Mish [15] activations predicts the noise; every residual block receives a FiLM scale and bias [17] from the diffusion-timestep embedding and the condition. The architecture is adapted from robomimic's Diffusion Policy implementation, scaled down to match BC's parameter budget and trained with the same optimizer

Table 1: Level 1 policy families. Parameter counts are exact; all four lie within the budget rule of the protocol (diffusion's count is not more than twice BC-MLP's, so no capacity-matched BC variants were added).

Family	Network	Horizons	Parameters
BC-MLP	MLP 2 × 1024, ReLU	$T_o = 1$, $T_p = 1$, $T_a = 1$	1,077,255
BC-RNN	LSTM 2 × 296	context 10, $T_p = 1$, $T_a = 1$	1,080,703
BC-MLP-chunk	MLP 2 × 1024, ReLU	$T_o = 2$, $T_p = 16$, $T_a = 8$	1,204,336
Diffusion U-Net	1D U-Net (64, 128), FiLM; DDPM $K = 100$, DDIM-10, EMA	$T_o = 2$, $T_p = 16$, $T_a = 8$	1,092,551

recipe as BC rather than robomimic's warm-up schedule; its scheduler semantics were checked against diffusers [25].

4.2 Diffusion training and sampling

Training uses the DDPM forward process [9] with $K = 100$ steps and the squared-cosine noise schedule [16]: $A_t = \sqrt{\bar{\alpha}_t}\, A_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon$ for a normalized demonstrated chunk A_0, a timestep t drawn uniformly and Gaussian noise ϵ. The network predicts ϵ and the loss is the masked squared error over the valid (unpadded) action steps. At inference the policy runs DDIM [23] with 10 steps and $\eta = 0$, so a chunk is a deterministic function of its initial noise; the predicted clean chunk is clipped to $[-1, 1]$ at every step. In min–max normalized space this keeps every diffusion action within the range of the training actions on each dimension, a bound the BC heads do not have (their outputs are clipped only to the controller limits; Section 9). The evaluated network is an exponential moving average (EMA) of the weights with power 0.75; the raw trained weights are kept in every checkpoint for ablation A1. DDIM with 10 steps was fixed before any reported run because the robomimic default (DDPM with all 100 steps) made every canonical evaluation several times longer; A2 compares the two.

4.3 Optimization and matched budgets

The base configuration of every family uses AdamW [13] with learning rate 1×10^{-4}, batch 100 and no gradient clipping; the tuning trials below change only the optimizer. Every main run trains for 100,000 gradient steps and saves a checkpoint every 5,000 steps (20 checkpoints), identically for all families and both tasks, so every family sees the same number of training samples. Training ran on the local CPU with one torch thread per run.

Tuning budget. Each family received exactly 4 configurations per task, each trained with the first training seed at the main budget and declared before the first run: the pilot configuration; a cosine learning-rate schedule annealed to zero; a constant learning rate of 3×10^{-4}; and weight decay 0.1. The architecture, horizons, batch and budget never changed. The configuration whose selected checkpoint had the most validation successes (ties: lower mean first-success step, then the lower trial number) was then trained with the other two seeds. The chosen trials were, on Lift, 4 (BC-MLP), 4 (BC-RNN), 2 (BC-MLP-chunk) and 3 (diffusion), and on Can 2, 2, 4 and 2.

4.4 Observation-modality extensions

Two further studies replace the privileged state by sensor observations on Lift, each with its own freeze. Both keep the diffusion head, horizons and action encoding above, train for 60,000 steps with batch 64 and 12 checkpoints on one NVIDIA L4 GPU with deterministic CUDA kernels, and evaluate on the same CPU platform as Level 1.

- **RGB study.** Two cameras (the front `agentview` and the wrist `robot0_eye_in_hand`) at 84×84 pixels, rendered from the recorded demonstration states, each randomly cropped during training (centre crop at evaluation) and encoded by a ResNet-18 [7] with GroupNorm and random initialization followed by a spatial soft-argmax [12], plus 9-dimensional proprioception (end-effector pose and gripper; no object state). Visual diffusion is compared with visual BC-chunk, which has the same encoder and horizons and a deterministic head.
- **Level 2 point clouds (DP3-inspired).** One world-frame cloud per step, fused from the depth images of the same two cameras, cropped to a fixed workspace box and reduced to 512 points by deterministic farthest-point sampling. A per-point MLP (widths 64 to 256, LayerNorm, ReLU) with max pooling and a 64-dimensional projection encodes it, following the idea of DP3 [28] and PointNet [19]; the same

proprioception is added. The cloud uses only depth, camera intrinsics and camera poses (the wrist camera's from robot kinematics). No segmentation, object pose or simulator label enters it, so the arm is sensor-like rather than privileged. The depth is noise-free and the calibration is exact. The reference arm is the RGB study's diffusion arm, frozen and never retrained.

5 Evaluation protocol

The protocol, its hypotheses and its analysis were committed before the runs they govern; every later amendment is dated in the repository, and no amendment changed an analysis method after its data existed. Three analysis scripts received input-reading fixes after their data existed (Section 9; Appendix A).

Initial-state suites. Each task has disjoint suites of 50 serialized initial states drawn from robosuite's standard reset distribution with reserved seed blocks: a *validation* suite, which selects every checkpoint and configuration; a *held-out test* suite, locked in code until a freeze document was committed and then run once per selected checkpoint; and a fresh *robustness base* suite (seeds from 300000) for the perturbation study. No two suites share a state or a seed.

Hypotheses. H1: every trained family beats a uniform random-action baseline. H2 (primary): diffusion versus BC-MLP and versus BC-RNN. H2-c: diffusion versus BC-MLP-chunk, required before any diffusion advantage over BC-MLP is attributed to the generative model rather than to chunking. H3: observation history, BC-RNN versus BC-MLP (and, in ablation A3, diffusion with one versus two observations). H4: robustness under two implemented perturbations (Section 7). H5 belongs to the separately frozen point-cloud study (Section 8) and is never pooled with H1 to H4. Each task has 8 declared success differences.

Selection. Each training run's checkpoint is the one with the most validation successes; ties go to the lower mean first-success step and then to the earlier checkpoint. The same rule chose each family's tuning trial. Because the validation suite chose among 20 checkpoints and 4 trials, validation success is biased upward; it is never reported as a test result (Table 15 in the appendix compares it with success on fresh and held-out starts).

Rollouts. Every start runs once ($R = 1$) with policy seed 0. For the diffusion family the initial noise of each decision is a deterministic function of the start, the policy seed and the decision index, so a rerun reproduces every action. With $R = 1$ the difficulty of a start and the diffusion noise drawn on it cannot be separated (Section 9).

Statistics. The primary metric is the success rate. Families are compared by a hierarchical paired bootstrap [4, 5] with $B = 10{,}000$ replicates and seed 20261106: each replicate resamples the 3 training seeds of each family with replacement, then resamples the 50 starts once and shares them across families, so every difference is paired by start. A difference is *separated* when its 95 % percentile interval excludes zero, and *not separated* otherwise; not separated is not evidence of equality. Intervals are not adjusted for multiplicity, and the counts of declared comparisons are reported with each analysis. Per-checkpoint rates carry Wilson intervals [26]. The secondary metric is the censored episode length: the first-success step, or the horizon 400 for a failure. Because a failure contributes the full horizon, a difference in censored length mixes a difference in success with a difference in speed; we decompose it in Section 6.

Sensitivity analysis (not preregistered). With 3 seeds per family the seed component of each bootstrap interval rests on very few values. After the audit (Section 9) we recomputed every declared difference with a seed-level t interval on the frozen per-seed rates: Welch's interval for separately trained families, a paired t interval for the same checkpoints under two conditions, and Welch's interval on the per-seed drops for a difference of drops. It treats per-seed means as the units, ignores pairing by start, and is undefined when the seeds do not vary. It was not preregistered, it changes no frozen verdict, and it is reported beside the frozen result as a check of how much each verdict depends on the method.

Failure taxonomy. Every failed episode is assigned, by a rule fixed before the test and computed from recorded states, to one of seven failure categories: engineering error, gripper never closed, no grasp with the gripper near the object, no grasp after drifting away, object dropped, holding the object at timeout, and (Can) object in the goal bin without success.

Table 2: Held-out success (%) per family: successes over episodes, the mean over the three training seeds with the hierarchical bootstrap's 95 % interval, and the per-seed rates.

Family	Lift k/n	Lift success %	Lift per seed	Can k/n	Can success %	Can per seed
BC-MLP	136/150	90.7 [83.3, 96.7]	88, 92, 92	115/150	76.7 [62.7, 90.0]	68, 88, 74
BC-RNN	150/150	100.0 [100.0, 100.0]	100, 100, 100	144/150	96.0 [89.3, 100.0]	92, 100, 96
BC-MLP-chunk	147/150	98.0 [94.0, 100.0]	98, 100, 96	144/150	96.0 [88.7, 100.0]	96, 92, 100
Diffusion U-Net	146/150	97.3 [93.3, 100.0]	98, 96, 98	143/150	95.3 [89.3, 100.0]	92, 98, 96
Random (uniform)	0/50	0.0 [0.0, 0.0]	0	0/50	0.0 [0.0, 0.0]	0

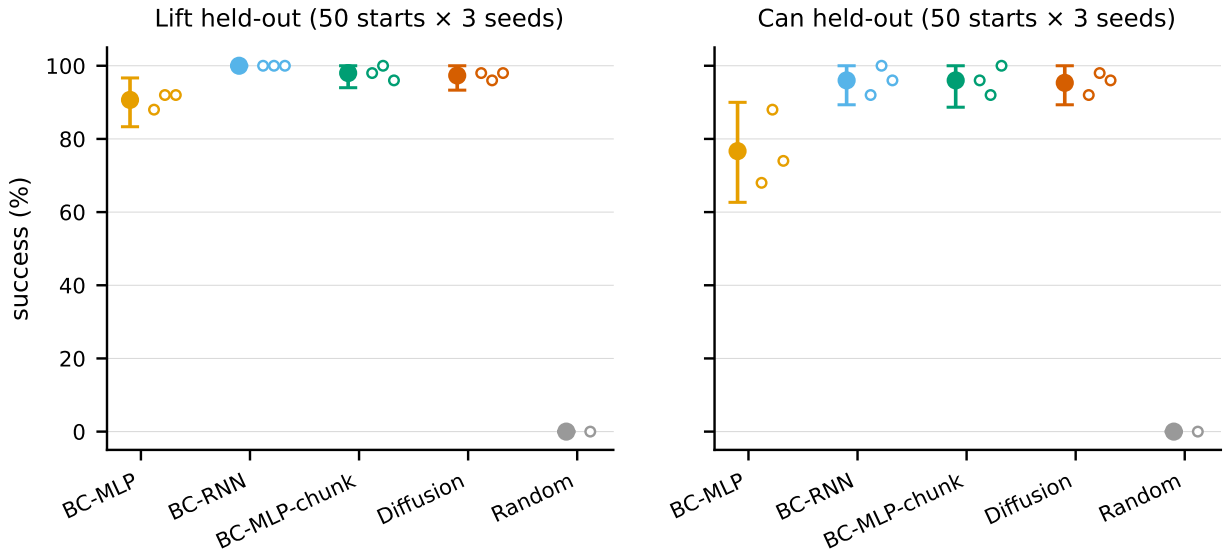


Figure 1: Held-out success per family and task. Filled markers: mean over seeds with the bootstrap 95 % interval; open markers: the three training seeds.

6 Held-out results (Level 1)

Table 2 and Figure 1 give the held-out success of every family; Table 3 gives every declared difference, and Figure 2 plots the comparisons that bear on H2, H2-c and H3. Each family ran 50 held-out starts per training seed on each task, once, after the Level 1 freeze (commit `bf0a0fa`).

H1. Every trained family separates from the random-action baseline, which never succeeds, on both tasks (Table 3).

Lift. Lift is close to saturated: BC-RNN succeeds on every held-out start (100.0 %), BC-MLP-chunk on 98.0 % and diffusion on 97.3 %. The only separated learned comparison is H3: BC-RNN exceeds BC-MLP by +9.3 pp [+3.3, +16.7]. Diffusion's advantage over BC-MLP (+6.7 pp [−0.7, +14.7]) is not separated, although the declared secondary outcome, censored episode length, is (−28.9 steps [−59.4, −1.6], mostly from success), nor is its difference from BC-RNN (−2.7 pp) or from BC-MLP-chunk (−0.7 pp).

Can. On Can, BC-MLP is the clear outlier (76.7 %), while the other three families lie within one point of each other (96.0, 96.0 and 95.3 %). The primary H2 comparison is separated: diffusion exceeds BC-MLP by +18.7 pp [+4.7, +32.7]. The H2-c control does not support attributing that advantage to the generative model: diffusion minus BC-MLP-chunk is −0.7 pp [−8.7, +7.3], is not separated (which is not evidence of equality). BC-RNN also separates from BC-MLP (H3: +19.3 pp [+5.3, +34.0]), although it differs from BC-MLP in its architecture as well as in its history. With privileged state on these two tasks, these data therefore give no evidence that the generative head adds to observation history and chunking. That BC-MLP-chunk (96.0 %) also exceeds BC-MLP (76.7 %) is visible in Table 2, but it is not a declared comparison and is exploratory. A further difference between the heads, the diffusion sampler's bound on its actions, is discussed in Section 9.

Table 3: Declared held-out differences $A - B$ with the hierarchical paired bootstrap's 95 % interval. Success in percentage points; censored episode length in control steps, with failures counted at the horizon (negative is faster). Intervals are not adjusted for the 8 success comparisons per task.

	Difference $A - B$	Lift	Can
Success (percentage points)			
H1	BC-MLP − Random (uniform)	+90.7 [+83.3, +96.7], separated (+)	+76.7 [+62.7, +90.0], separated (+)
H1	BC-RNN − Random (uniform)	+100.0 [+100.0, +100.0], separated (+)	+96.0 [+89.3, +100.0], separated (+)
H1	BC-MLP-chunk − Random (uniform)	+98.0 [+94.0, +100.0], separated (+)	+96.0 [+88.7, +100.0], separated (+)
H1	Diffusion U-Net − Random (uniform)	+97.3 [+93.3, +100.0], separated (+)	+95.3 [+89.3, +100.0], separated (+)
H2	Diffusion U-Net − BC-MLP	+6.7 [−0.7, +14.7], not separated	+18.7 [+4.7, +32.7], separated (+)
H2	Diffusion U-Net − BC-RNN	−2.7 [−6.7, +0.0], not separated	−0.7 [−8.0, +6.7], not separated
H2-c	Diffusion U-Net − BC-MLP-chunk	−0.7 [−5.3, +4.7], not separated	−0.7 [−8.7, +7.3], not separated
H3	BC-RNN − BC-MLP	+9.3 [+3.3, +16.7], separated (+)	+19.3 [+5.3, +34.0], separated (+)
Censored episode length (steps; failures count 400)			
H2	Diffusion U-Net − BC-MLP	−28.9 [−59.4, −1.6], separated (−)	−56.0 [−98.0, −14.7], separated (−)
H2	Diffusion U-Net − BC-RNN	+11.0 [+1.1, +25.1], separated (+)	+2.1 [−18.5, +22.7], not separated
H2-c	Diffusion U-Net − BC-MLP-chunk	+2.8 [−15.6, +20.0], not separated	+9.5 [−13.0, +33.3], not separated
H3	BC-RNN − BC-MLP	−39.9 [−67.8, −16.2], separated (−)	−58.1 [−100.3, −16.7], separated (−)

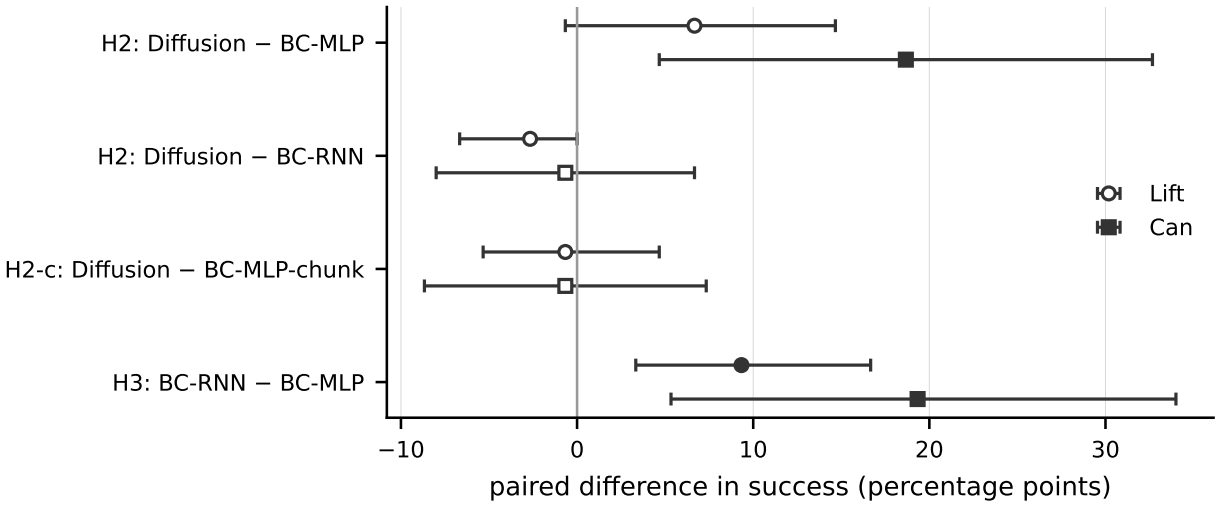


Figure 2: The H2, H2-c and H3 differences in held-out success with their 95 % intervals. Filled markers are separated (the interval excludes zero).

How much the verdicts depend on the interval method. The seed-level sensitivity analysis (Section 5; Table 17; not preregistered) changes 3 of the 16 Level 1 verdicts. Both separated Can comparisons among learned families lose separation: diffusion minus BC-MLP becomes [−4.5, +41.8] with 2.4 degrees of freedom, and BC-RNN minus BC-MLP [−2.8, +41.5], because BC-MLP's three Can seeds differ widely (Table 2). Lift's diffusion minus BC-MLP, not separated by the bootstrap, becomes separated ([+1.9, +11.5]). The frozen verdicts stand as the preregistered result, but the Can H2 and H3 separations rest on three BC-MLP seeds and should be read as fragile. No H2-c verdict changes under either method.

Censored episode length. Length differences are dominated by failures, which count 400 steps. We split each difference exactly into a success part, $-(p_A - p_B)(H - \bar{T}_B)$, and a speed part, $p_A(\bar{T}_A - \bar{T}_B)$, where p is the success rate, $\bar{T}$ the mean first-success step of successes and H the horizon (D-095; not preregistered). Diffusion's separated length advantage over BC-MLP on Can (−56.0 steps) is almost all success (−52.6 from success, −3.4 from speed). Its separated length *disadvantage* against BC-RNN on Lift (+11.0 steps) is also mostly success (+9.5 from diffusion's few failures, against +1.5 from speed). Censored length therefore restates the success comparison and adds little independent evidence about speed.

Selected checkpoints. The selected training steps are listed per run in Table 14. Of the 18 BC runs, 4 selected their first checkpoint, so for them the budget was more than enough and earlier checkpoints were not examined; two diffusion runs (the first Lift seed and the second Can seed) selected their final checkpoint (100,000 and 100,000 steps), so a longer budget might have helped them.

Table 4: Declared secondary metrics on the held-out starts, aggregated after the frozen analysis (D-096; plain means, no intervals). Gripper sign changes per episode, with the range of the per-seed means; mean absolute step-to-step change of the executed motion command; and the clip fraction, the frozen mean over the six motion dimensions of the share of actions clipped to the controller limits. All depend on episode length, which ends at first success.

Task	Family	Episodes	Gripper sign changes	Mean $\lvert\Delta a\rvert$ (motion)	Clipped motion actions (%)
Lift	BC-MLP	150	3.06 [1.46, 5.56]	0.017	15.7
Lift	BC-RNN	150	1.03 [1.00, 1.08]	0.019	0.0
Lift	BC-MLP-chunk	150	2.30 [1.00, 3.60]	0.026	4.5
Lift	Diffusion U-Net	150	2.31 [1.90, 3.02]	0.036	0.0
Lift	Random (uniform)	50	197.78	0.670	0.0
Can	BC-MLP	150	6.04 [3.24, 9.02]	0.020	9.5
Can	BC-RNN	150	1.97 [1.96, 2.00]	0.022	2.4
Can	BC-MLP-chunk	150	2.73 [2.00, 3.96]	0.025	3.8
Can	Diffusion U-Net	150	2.57 [2.42, 2.70]	0.043	0.0
Can	Random (uniform)	50	198.76	0.666	0.0

Secondary metrics. Table 4 reports the protocol's declared secondary metrics, which every held-out summary computed per episode but no frozen analysis aggregated (audit C-9; D-096). They are descriptive only, without intervals. The mean absolute step-to-step change of the executed motion command is 0.036 for diffusion and 0.026 for BC-MLP-chunk on Lift; BC-MLP has the largest clip fraction (15.7 % on Lift), and diffusion's is zero by construction of its sampler (Section 9).

Cost and latency. Table 5 lists training cost and inference latency. On the CPU, a diffusion run costs 19–26 times the CPU hours of a BC-MLP-chunk run, and a diffusion decision (10 DDIM steps) takes 22–27 times as long, for no measured gain in held-out success over that control (it did gain under the Lift arm offset; Section 7). Every Level 1 run trained locally at no billed cost. The table's latencies come from two methods (a dedicated benchmark for Level 1, medians recorded during the held-out rollouts for the GPU-trained studies) and are comparable only within a block. The step time of the first Level 2 training seed printed by the frozen analysis was wrong; the table shows the frozen value beside the corrected one (KI-049). Table 6 reconciles all paid compute: 24 launches, $10.61 in total, each under a committed approval and within its dollar cap; one approval's launch count was exceeded before it was amended (Section 9).

7 Robustness and ablations (Level 1)

Design. H4 tests two perturbations defined, validated and committed before the first robustness rollout. *Cube shift* (Lift only) moves the cube's centre to a radius drawn uniformly between 5 and 7 cm from the table centre, at a uniform angle; the demonstrations place it within ±3 cm in each direction, so every shifted cube lies outside the training square. *Arm offset* (Lift and Can) adds ±0.10 rad with independent random signs to every arm joint, 5 times the standard deviation (0.02 rad) of robosuite's own joint noise. Can has no placement set because its can is already sampled over the whole source bin. Each perturbed state is derived from the base state with the same index and passes validity checks (exact restore, no success at the start, joint limits, no new contacts or penetration, and passive settling). The 3 selected checkpoints of each family run every base and perturbed start once. Since the base starts selected nothing, these results are not biased by selection, but they were run before the held-out test and are not part of it. Differences are paired by base index, and the bootstrap is the one of Section 5.

Lift. All four families succeed on at least 98.0 % of the base starts (Table 7, Figure 4). Under cube shift, BC-MLP and BC-MLP-chunk fall to 52.7 and 54.7 %, diffusion to 85.3 %, and BC-RNN holds at 96.0 %. Under arm offset, diffusion keeps 88.0 %, against 50.0 % for BC-RNN, 48.7 % for BC-MLP-chunk and 22.7 % for BC-MLP. Diffusion exceeds BC-MLP-chunk on both sets (+30.7 and +39.3 pp, both separated) and BC-RNN under arm offset (+38.0 pp, separated), while BC-RNN is ahead under cube shift (diffusion minus BC-RNN −10.7 pp, not separated).

Can. The Lift pattern does not carry over. Under arm offset on Can, BC-MLP-chunk keeps 79.3 % and diffusion 71.3 %; their difference (−8.0 pp [−23.3, +8.7]) is not separated. Diffusion does exceed BC-MLP (+41.3 pp) and BC-RNN (+18.0 pp).

Table 5: Training cost and inference latency. Level 1 CPU hours are per training seed; latency is the median time of one decision (observation to action or action chunk); chunked policies decide every 8 control steps. Billed amounts are Modal's usage report at list price before credits.

Run	Hardware and training time	Latency ms	ms/step	Billed
Level 1: training on local CPU (one torch thread); latency: dedicated benchmark, canonical container				
Lift BC-MLP	CPU 0.22, 0.17, 0.17 CPU-h	0.91	–	$0.00
Lift BC-RNN	CPU 1.20, 0.93, 0.93 CPU-h	9.04	–	$0.00
Lift BC-MLP-chunk	CPU 0.17, 0.18, 0.18 CPU-h	1.66	–	$0.00
Lift Diffusion U-Net	CPU 4.42, 3.36, 3.36 CPU-h	36.95	–	$0.00
Can BC-MLP	CPU 0.22, 0.16, 0.16 CPU-h	1.08	–	$0.00
Can BC-RNN	CPU 1.27, 0.85, 0.85 CPU-h	9.00	–	$0.00
Can BC-MLP-chunk	CPU 0.18, 0.18, 0.18 CPU-h	1.42	–	$0.00
Can Diffusion U-Net	CPU 4.31, 3.99, 3.98 CPU-h	37.71	–	$0.00
RGB study (Lift): training on one NVIDIA L4; latency: median of episode medians during the held-out run				
Visual BC-chunk seed 0	L4 1.01 h, 2 segments	66.4	–	$1.14
Visual BC-chunk seed 1	L4 1.01 h, 2 segments	65.0	–	$1.14
Visual BC-chunk seed 2	L4 1.00 h, 2 segments	63.0	–	$1.13
Visual diffusion seed 0	L4 1.35 h, 2 segments	183.1	–	$1.52
Visual diffusion seed 1	L4 1.36 h, 2 segments	187.2	–	$1.53
Visual diffusion seed 2	L4 1.29[b] h, 3 segments	166.4	–	$1.60
Level 2 (Lift): training on one NVIDIA L4; latency: median of episode medians during the held-out run				
Point-cloud diffusion seed 0	L4 0.41[a] h, 3 segments	147.4	24.62 → **42.2**[a]	$0.80
Point-cloud diffusion seed 1	L4 0.70 h, 1 segment	146.0	41.73	$0.78
Point-cloud diffusion seed 2	L4 0.71 h, 1 segment	147.0	42.55	$0.80

[a] Erratum (KI-049, `ERRATA.md`, SHA-256 `280974a9`): the frozen analysis printed the left value. The corrected step time is the completing launch's training time over the steps it trained, from its committed launch record; the hours are that segment only and leave out the cut first segment (`ERRATA.md` gives the total).

[b] One launch of this run was preempted and restarted; the preempted attempt recorded no training time, so the hours leave it out, while the billed amount includes it (KI-044; audit C-6).

Table 6: Paid compute by approval, reconciled with the cost ledger's billing events.

Approval	Gate	Functions	Billed launches	Cap	Billed
A-01	Gate A	`gpu_probe`, `train_cpu`	6	$2.00	$0.1483
A-02	Gate D (RGB probe)	`rgb_probe`	1	$1.00	$0.0317
A-03	Gate D2 (RGB main)	`train_gpu`	12	$15.00	$8.0598
A-04	Level 2 (point cloud)	`train_gpu_pc`	5	$9.00	$2.3747
Total			24		$10.6144

Reading H4. Each Lift and Can analysis reports 24 and 12 differences and contrasts, unadjusted for multiplicity. The seed-level sensitivity analysis (Table 18 in the appendix) changes 6 of 41 robustness verdicts, all from separated to not separated. Among them is diffusion's advantage over BC-MLP-chunk under Lift cube shift ([−18.9, +80.2]), whereas its advantage under Lift arm offset stays separated under both methods. Two further cautions apply. First, the diffusion sampler bounds every action to the range of the training actions, while the BC heads are bounded only by the controller limits. In BC episodes that fail under perturbation most motion actions lie outside that range, which diffusion cannot do, so the between-family robustness gaps combine the generative model with this output bound (Section 9). This is a correlation, not a mechanism: the out-of-range actions start only after a grasp is missed, and BC-RNN rarely leaves the range yet keeps only 50.0 % under arm offset. Second, the results hold only for these two perturbations. We therefore read H4 as: on Lift, diffusion degrades least under a large joint offset, and under cube shift BC-RNN degrades least, with diffusion not separated from it; on Can, diffusion's drop under arm offset is not separated from the chunked MLP's (+4.7 pp [−10.7, +20.7]), whose point estimate is smaller.

Ablations. Table 8 reports the three preregistered diffusion ablations on the robustness base starts. A1 evaluates the raw trained weights instead of the EMA weights of the same checkpoints; A2 samples with DDPM over all 100 steps instead of DDIM with 10; A3 (Can only) trains a diffusion policy with one observation instead of two (1,063,367 instead of 1,092,551 parameters, three new seeds, selected by the same rule). Checkpoints were selected with EMA weights and DDIM-10, which favours the primary arm in A1 and A2,

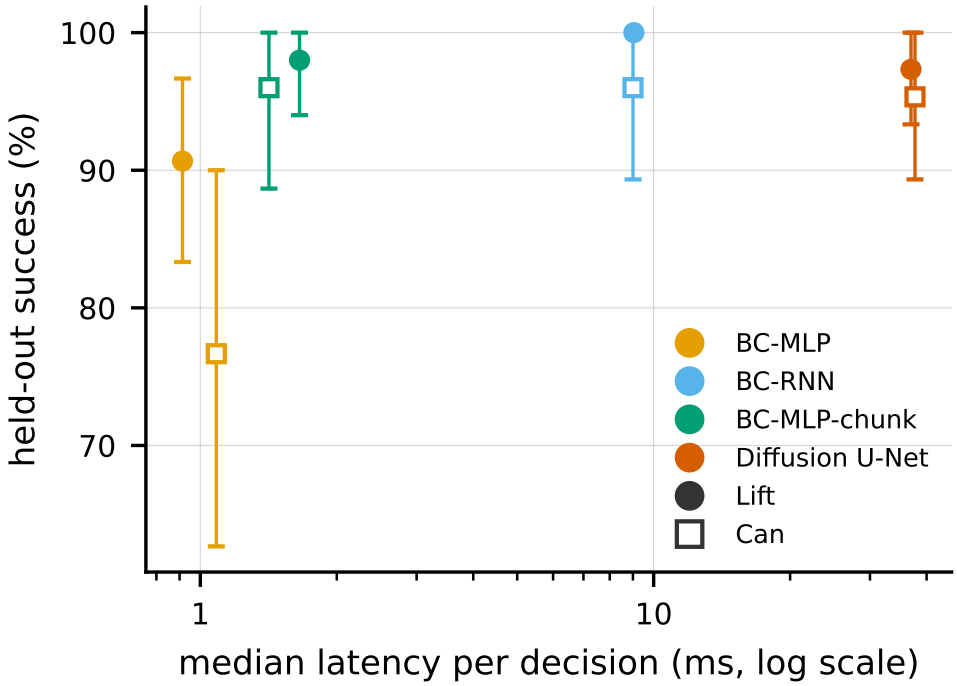


Figure 3: Held-out success against the median latency of one decision (log scale) for the Level 1 families, with the bootstrap interval. Filled circles: Lift; open squares: Can.

Table 7: Success (%) on the robustness base starts and the two perturbation sets, mean over seeds with the bootstrap 95 % interval.

Family	Lift base	Lift cube shift	Lift arm offset	Can base	Can arm offset
BC-MLP	98.0 [92.0, 100.0]	52.7 [35.3, 69.3]	22.7 [12.0, 34.0]	69.3 [56.0, 82.0]	30.0 [19.3, 41.3]
BC-RNN	100.0 [100.0, 100.0]	96.0 [88.0, 100.0]	50.0 [36.0, 64.0]	96.7 [91.3, 100.0]	53.3 [41.3, 66.0]
BC-MLP-chunk	100.0 [100.0, 100.0]	54.7 [33.3, 75.3]	48.7 [36.0, 61.3]	95.3 [89.3, 100.0]	79.3 [66.0, 90.7]
Diffusion U-Net	99.3 [96.0, 100.0]	85.3 [76.0, 93.3]	88.0 [79.3, 95.3]	92.0 [85.3, 98.0]	71.3 [58.0, 84.0]

and all three ablations share the primary arm's evaluations on the base starts, so they are not independent of each other or of H4. None of the five differences is separated, by the bootstrap or by the seed-level analysis. DDIM-10 was not separated from DDPM over all 100 steps, although both point estimates favour DDPM (-0.7 and -3.3 pp). No ablation of the action horizons (T_p, T_a) was run; the only comparison that changes them is BC-MLP against BC-MLP-chunk, which also changes the observation window and is not a declared comparison.

Failure modes. Table 9 and Figure 5 classify every held-out and Level 2 failure, and Table 10 every failure of the robustness evaluations. Under Lift arm offset, BC-MLP's failures are mostly drifts (104 of its episodes), while diffusion has 0 drifts and fails mostly next to the cube (12). On the Level 1 held-out starts, BC-MLP's Lift failures are mostly drifts away from the cube without a grasp (12 of 50×3 episodes), while diffusion's few failures end near the cube or after a drop. On Can, BC-MLP fails in several ways, most often by never closing the gripper (11 episodes). The point-cloud policy's failures under cube shift are mostly misses near the cube (45) and drops (16), not drifts.

8 Observation-modality extensions (Lift)

Both extensions replace the privileged object pose by sensor observations, each under its own freeze committed before its first training launch. They share Lift's demonstrations, split, proprioception, action encoding, diffusion head and horizons. Neither is compared causally with Level 1: the Level 1 families trained longer on a CPU, received 4 tuning trials each (the extensions received none), and saw the exact object pose, so a Level 1 number placed beside an extension's number on the same starts is context, not a comparison. Figure 6 shows the scenes and the point cloud.

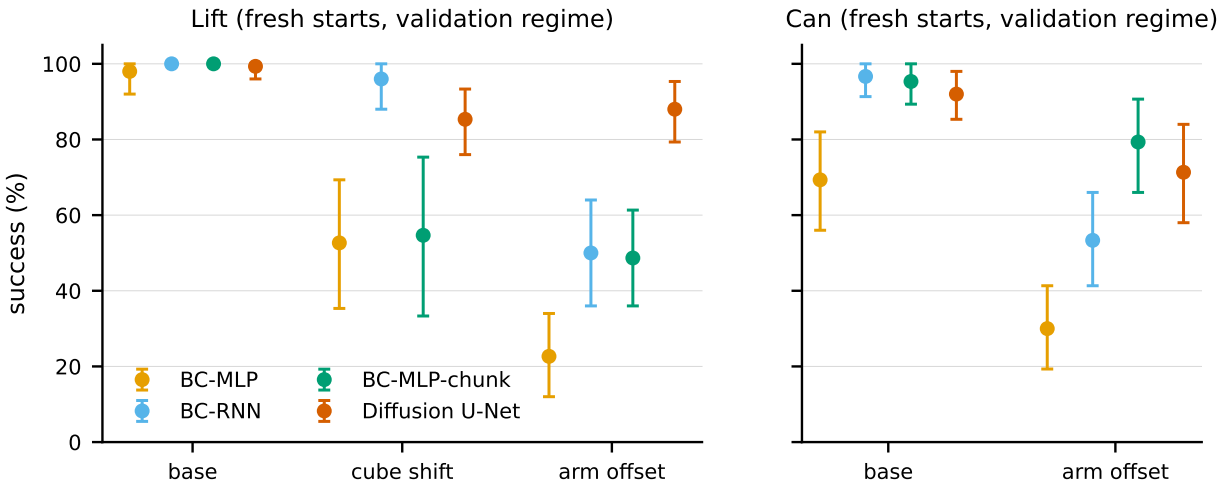


Figure 4: Success per family on the base starts and under each perturbation (mean over seeds with the 95 % interval).

Table 8: Diffusion ablations on the robustness base starts (3 seeds, 50 starts each). A1 and A2 re-evaluate the selected checkpoints; A3 trains separately. Primary: EMA weights, DDIM-10, $T_o = 2$.

	Task	Arm A (success %)	Arm B (success %)	$A - B$ (pp)	Outcome
A1	Lift	primary 99.3 [96.0, 100.0]	trained weights 98.7 [95.3, 100.0]	+0.7 [+0.0, +2.7]	not separated
A1	Can	primary 92.0 [84.7, 98.0]	trained weights 94.7 [88.7, 99.3]	−2.7 [−7.3, +0.7]	not separated
A2	Lift	primary 99.3 [96.0, 100.0]	DDPM-100 100.0 [100.0, 100.0]	−0.7 [−4.0, +0.0]	not separated
A2	Can	primary 92.0 [84.7, 98.0]	DDPM-100 95.3 [89.3, 100.0]	−3.3 [−10.0, +3.3]	not separated
A3	Can	$T_o = 2$ 92.0 [84.7, 98.0]	$T_o = 1$ 94.0 [86.7, 99.3]	−2.0 [−8.7, +4.0]	not separated

8.1 RGB study: visual BC versus visual diffusion

Table 11 gives the result. Both arms are saturated on the Lift held-out starts: visual diffusion succeeds on 150 of 150 episodes and visual BC-chunk on 149. The preregistered difference, +0.7 pp [+0.0, +4.0], is not separated, and the seed-level interval agrees ([−2.2, +3.5]). With both arms at or near 50 of 50 per seed, this test cannot distinguish them; a harder task or set would be needed. Under saturation, checkpoint selection was decided mostly by sub-step differences in mean first-success step among checkpoints that all succeeded on every validation start, so it carried little information.

8.2 Level 2: a DP3-inspired point-cloud policy (H5)

Question and design. H5 asks whether the point-cloud diffusion policy succeeds more often than the frozen RGB diffusion policy when the cube is placed outside the demonstrations' placement square. The primary set is Lift cube shift, paired by start with the robustness base starts (Section 7); neither arm selected anything on either set. The set was chosen after the Level 1 robustness and RGB held-out results were known, because the RGB held-out test was saturated and cube shift was the one implemented perturbation that leaves the training placement; the point-cloud arm's design was fixed in its freeze before any of its rollouts. The arm is DP3-inspired in its encoder only (Section 4); it is not a reproduction of DP3.

Result. The answer is no, in the opposite direction (Table 12, Figure 7). Both arms succeed on every base start. Under cube shift the point-cloud arm falls to 56.7 % while the RGB arm keeps 98.0 %; H5's difference is −41.3 pp [−54.0, −28.0], separated, and the seed-level interval agrees ([−57.3, −25.4]). The point-cloud arm's own drop (+43.3 pp) is separated and the RGB arm's (+2.0 pp) is not. Because both arms are at the ceiling on the base starts, the difference of drops equals H5 with its sign reversed and adds no information. The point-cloud arm's failures are mostly misses next to the displaced cube and drops (Table 9). On the held-out starts both arms succeed on every episode, so that secondary comparison is uninformative; it is a check that the point-cloud arm works in distribution, not evidence of equivalence.

What the result does and does not show. H5 compares two complete policies, not two modalities. Besides the input, the arms differ in their encoders (a ResNet-18 per camera, 22,394,176 encoder parameters

Table 9: Failure taxonomy: episodes per category, summed over the three seeds (one run of 50 episodes for the random baseline).

Policy	success	engineering error	gripper never closed	no grasp, drifted	no grasp, near	timeout holding	in goal bin, not success	dropped
Level 1 held-out, Lift								
BC-MLP	136	0	0	12	0	0	0	2
BC-RNN	150	0	0	0	0	0	0	0
BC-MLP-chunk	147	0	0	3	0	0	0	0
Diffusion U-Net	146	0	0	0	3	0	0	1
Random (uniform)	0	0	0	38	9	0	0	3
Level 1 held-out, Can								
BC-MLP	115	0	11	8	4	3	2	7
BC-RNN	144	0	1	1	0	0	3	1
BC-MLP-chunk	144	0	1	0	4	0	1	0
Diffusion U-Net	143	0	0	2	0	0	3	2
Random (uniform)	0	0	0	44	6	0	0	0
RGB study held-out, Lift								
Visual diffusion	150	0	0	0	0	0	0	0
Visual BC-chunk	149	0	0	0	1	0	0	0
Level 2, Lift								
Point-cloud diffusion, held-out	150	0	0	0	0	0	0	0
Point-cloud diffusion, cube shift	85	0	0	4	45	0	0	16
Point-cloud diffusion, base	150	0	0	0	0	0	0	0
Visual diffusion, cube shift	147	0	0	0	3	0	0	0
Visual diffusion, base	150	0	0	0	0	0	0	0

in total, against 357,184 for the point encoder), in augmentation (random crops for RGB, none for the cloud), and in the frame of the input: the cloud is in absolute world coordinates, so shifting the cube moves its points to coordinates the encoder never saw during training, whereas a convolutional encoder with random crops may be more tolerant of translation (not tested here). The depth is noise-free and the calibration exact. The result is therefore that *this* point-cloud package, trained on 180 demonstrations without augmentation, generalizes worse to unseen cube positions than the matched RGB package. It does not test, and should not be read as contradicting, the generalization results reported for DP3 [28] under its own design and tasks.

9 Limitations and threats to validity

An independent audit of the code and the frozen files, run after every result was frozen, produced the findings below; the full list, with evidence and the disposition of each, is in `docs/AUDIT.md`. No finding changed a frozen file. Where a finding could be addressed from committed data without a new rollout, it is addressed in this report and labelled.

Few seeds, and verdicts that depend on the method. Each family has 3 training seeds. The seed-level sensitivity analysis (Tables 17 and 18) changes 3 of 16 Level 1 verdicts, 6 of 41 robustness and ablation verdicts, and 0 of 7 verdicts of the two extensions; it is undefined for 1, 1 and 2 of them because the seeds do not vary. The separated Can H2 and H3 results, and several robustness separations, are among the verdicts that change. The audit also found two verdicts that change with the bootstrap's random seed. Of the 64 differences in the sensitivity tables (the declared ones and one context row), 10 have a frozen interval with a bound at exactly zero or of zero width; with 50 binary outcomes per seed the bootstrap distribution is discrete, so these verdicts rest on ties. A policy that succeeds on all 50 starts has a Wilson interval of [92.9, 100.0] %, so near the ceiling the test separates only large differences.

Table 10: Failure taxonomy of the robustness evaluations (validation regime), summed over the three seeds.

Policy	success	engineering error	gripper never closed	no grasp, drifted	no grasp, near	timeout holding	in goal bin, not success	dropped
Lift, base								
BC-MLP	147	0	0	3	0	0	0	0
BC-RNN	150	0	0	0	0	0	0	0
BC-MLP-chunk	150	0	0	0	0	0	0	0
Diffusion U-Net	149	0	0	0	0	0	0	1
Lift, cube shift								
BC-MLP	79	0	5	57	1	0	0	8
BC-RNN	144	0	0	4	0	0	0	2
BC-MLP-chunk	82	0	0	49	7	0	0	12
Diffusion U-Net	128	0	0	1	15	0	0	6
Lift, arm offset								
BC-MLP	34	0	2	104	6	0	0	4
BC-RNN	75	0	7	57	5	0	0	6
BC-MLP-chunk	73	0	0	67	3	0	0	7
Diffusion U-Net	132	0	0	0	12	1	0	5
Can, base								
BC-MLP	104	0	16	9	4	2	8	7
BC-RNN	145	0	1	1	0	0	2	1
BC-MLP-chunk	143	0	2	1	1	0	1	2
Diffusion U-Net	138	0	0	0	2	1	4	5
Can, arm offset								
BC-MLP	45	0	17	60	13	0	6	9
BC-RNN	80	0	6	43	2	6	4	9
BC-MLP-chunk	119	0	5	8	9	0	3	6
Diffusion U-Net	107	0	8	7	8	0	4	16

Multiplicity. No interval is adjusted for multiplicity. Each Level 1 task has 8 success comparisons and 4 length comparisons, and the robustness analyses report 24 (Lift) and 12 (Can) differences and contrasts. A Bonferroni correction within each analysis would remove several separations, among them Can H2; the audit (`docs/AUDIT.md`, B-1 and B-6) gives the corrected intervals and the count.

The diffusion sampler bounds its actions. The sampler clips the predicted clean action chunk to $[-1, 1]$ in min–max normalized space, so every diffusion action lies within the range of the training actions on each dimension. The BC heads are clipped only to the controller limits, which are wider. The audit found that in BC episodes that fail under perturbation most motion actions lie outside the training range, while diffusion's never can. This is a correlation, not a shown mechanism: the out-of-range actions begin only after a grasp is missed, BC-MLP leaves the range on many base starts that it still solves, and BC-RNN rarely leaves it yet loses half its success under the Lift arm offset. This bound is part of what H2-c compares: diffusion and BC-MLP-chunk differ in the generative head *and* in this output bound, so neither the Level 1 nor the robustness differences between them isolate the generative model. Clamping BC outputs to the training range would test this without retraining; it was not run in this report, because it would be a new evaluation after the freeze.

Selection and its bias. Validation success, which chose among 20 checkpoints and 4 trials, is optimistic: Table 15 shows it beside success on the fresh base starts and on the held-out starts (for example BC-MLP on Can: 82.7 % on validation, 76.7 % held out). Of the 18 BC runs, 4 selected their first checkpoint, and two diffusion runs their last, so the checkpoint grid may have been too coarse for BC and the budget too short for those diffusion runs. In the RGB and Level 2 studies most checkpoints succeeded on every validation start, so selection there was decided by tiny differences in episode length and carries almost no information about the quantities those studies test; one RGB diffusion run and one point-cloud run selected their final

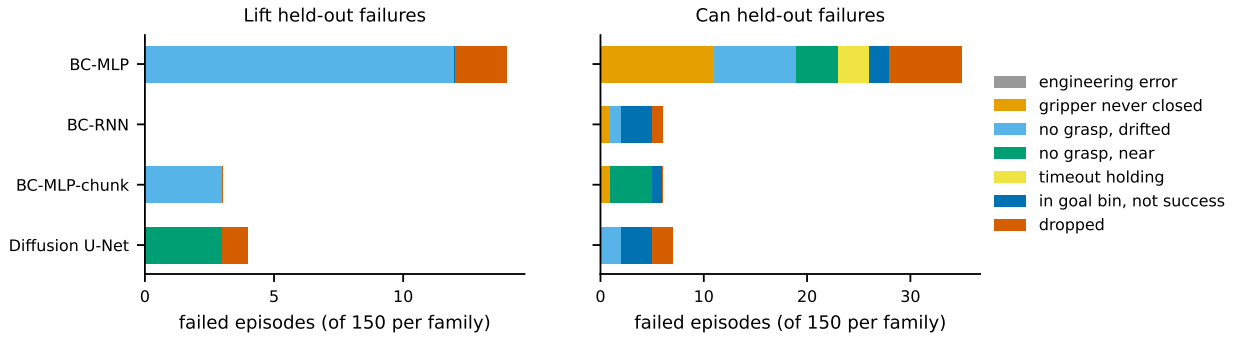


Figure 5: Failure categories of the Level 1 held-out episodes per family (successes omitted).



Figure 6: Illustrations, not selected for outcome. Left and centre: MuJoCo renders of recorded held-out states of the Lift and Can diffusion policies (first training seed, first held-out start; both episodes succeeded), regenerated from the stored simulator state. Right: the 512 sampled points of one Lift demonstration frame, projected onto the camera images.

checkpoint. Each family had only 4 tuning trials, and the extensions none, so no family is tuned to its best. Before the Lift search space was committed, a short pipeline rehearsal (a few hundred training steps) had evaluated the default and the higher-learning-rate configurations of BC-MLP and diffusion on the validation suite; the higher learning rate was later chosen for Lift diffusion. The trial list was the same for every family, but BC-RNN and BC-MLP-chunk had no such early look.

One rollout per start. With $R = 1$, a start's difficulty and the diffusion noise drawn on it are confounded, so diffusion's interval mixes policy stochasticity with start difficulty, unlike the deterministic BC families (KI-028).

Reuse of starts across studies. The same 50 Lift held-out starts served the Level 1 test, the RGB test and the Level 2 held-out check, and the robustness base and cube-shift starts served both H4 and H5. No model, checkpoint or configuration was ever selected on them. However, the three studies were designed in sequence and later designs were informed by earlier outcomes on these starts: H5's primary set was chosen after the Level 1 robustness and RGB held-out results were known. The Level 2 held-out comparison, with both arms at the ceiling, could not have favoured the point-cloud arm and is a replication-type check, not a test.

Scale of the diffusion policy. The diffusion family was matched to BC in parameters (about a million) and in its optimizer recipe. robomimic's default U-Net is wider and deeper and is trained with a warm-up cosine schedule, so these results speak to diffusion at this scale and recipe only.

In-distribution starts. The evaluation starts come from the same reset distribution as the demonstrations; on Lift each start lies within a few millimetres of a demonstration start (KI-011), so Lift held-out success measures in-distribution performance. Open-loop replay of the Can demonstrations diverges at contacts by several centimetres (KI-004), so the simulator reproduces the recorded dynamics only approximately.

Table 11: RGB study, Lift held-out: per-checkpoint results and the preregistered difference.

Family	Seed	Step	k/n	Wilson / bootstrap 95 %	First success / per seed
Visual BC-chunk	0	5,000	49/50	[89.5, 99.6]	42.8
Visual BC-chunk	1	5,000	50/50	[92.9, 100.0]	42.2
Visual BC-chunk	2	55,000	50/50	[92.9, 100.0]	44.7
Visual diffusion	0	55,000	50/50	[92.9, 100.0]	44.5
Visual diffusion	1	55,000	50/50	[92.9, 100.0]	44.3
Visual diffusion	2	60,000	50/50	[92.9, 100.0]	44.1
Visual BC-chunk	all			99.3 [96.0, 100.0]	98.0, 100.0, 100.0
Visual diffusion	all			100.0 [100.0, 100.0]	100.0, 100.0, 100.0
Visual diffusion − Visual BC-chunk: +0.7 pp [+0.0, +4.0], not separated					

Table 12: Level 2 results. Placement sets: success per arm, the declared contrasts and one context row (point cloud minus RGB on the base starts, not declared), with the hierarchical paired bootstrap's 95 % interval. Held-out: secondary.

Arm or contrast	Set	Success % or difference [95 % CI]	Per seed or outcome
Placement (validation regime; primary H5 on the cube shift)			
Point-cloud diffusion	base	100.0 [100.0, 100.0]	100.0, 100.0, 100.0
Point-cloud diffusion	cube shift	56.7 [43.3, 70.0]	64.0, 56.0, 50.0
Visual diffusion	base	100.0 [100.0, 100.0]	100.0, 100.0, 100.0
Visual diffusion	cube shift	98.0 [93.3, 100.0]	100.0, 98.0, 96.0
H5: point cloud − RGB on cube shift		−41.3 [−54.0, −28.0]	separated (−)
drop of Point-cloud diffusion (base − cube shift)		+43.3 [+30.0, +56.7]	separated (+)
drop of Visual diffusion (base − cube shift)		+2.0 [+0.0, +6.7]	not separated
point cloud − RGB on base		+0.0 [+0.0, +0.0]	not separated
drop point cloud − drop RGB		+41.3 [+28.0, +54.0]	separated (+)
Held-out (secondary; RGB records reused, never rerun)			
Point-cloud diffusion	held-out	100.0 [100.0, 100.0]	100.0, 100.0, 100.0
Visual diffusion	held-out	100.0 [100.0, 100.0]	100.0, 100.0, 100.0
point cloud − RGB, success		+0.0 [+0.0, +0.0]	not separated
point cloud − RGB, censored length (steps)		+1.1 [−1.4, +3.4]	not separated

Privilege, features and platform. Every Level 1 result uses exact object poses. The Can input recomputes the gripper-to-object vector from the current step, unlike robomimic's stored feature, so Can numbers are not comparable with robomimic's. The data are the robosuite v1.5 repackaging of the robomimic datasets, so no number here should be compared with the original benchmark. All rollouts ran in simulation on one platform. Our success predicate is evaluated on the current step's kinematics, while robosuite's own predicate reads the previous substep; in a small fraction of episodes ours fires one control step earlier, which shifts first-success steps but not success. The point cloud uses noise-free depth, exact camera calibration and a crop box whose lower edge removes the table at its known height.

Declared quantities not reported, and quantities not declared. The protocol's secondary metrics (gripper sign changes, action smoothness) were computed per episode but never aggregated by the frozen analysis; Table 4 reports them post hoc. The diffusion-specific plan-diversity metric was measured only in a pilot and is not reported, and the point-cloud throughput listed in the Level 2 freeze is not in its results. Validation success at every checkpoint, declared for the RGB study, and the point-cloud per-checkpoint rates listed in the Level 2 freeze are in the committed selection files but not in this report. The H4 analysis compares four of the six family pairs, a choice fixed in code before any robustness rollout. The optional DiT and GMM-BC families and the Square task were not run, to stay within the project's time and compute budget; the declared hypotheses did not depend on them.

Analysis code, records and cost figures. The Level 1 analysis script was hash-frozen; the robustness, RGB and Level 2 analysis scripts were not, and each received a small fix after its data existed. The fixes touched input reading and training-time accounting only, not pairing, resampling or which evaluations enter. The validation summaries of one RGB training run were recorded from a working tree with uncommitted changes; the next commit changed only documentation, logs and reports, and the frozen code hashes match. The Level 2 analysis misreported one training time (KI-049), and one RGB run's hours leave out a preempted

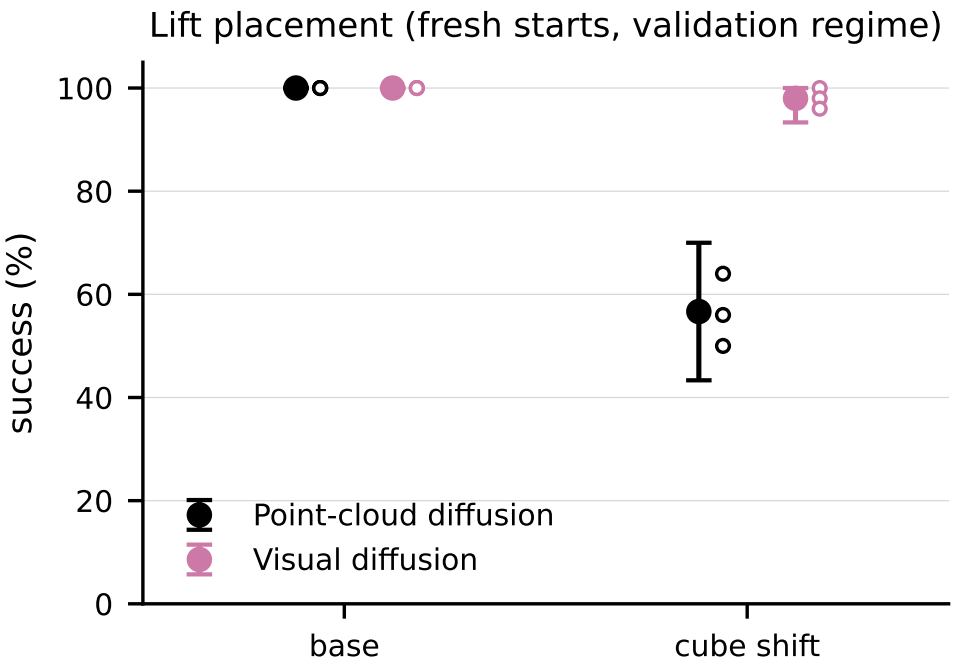


Figure 7: Level 2: success of the point-cloud and RGB diffusion arms on the base starts and under cube shift, with per-seed values.

attempt (Table 5). Billed amounts are Modal's usage report at list price before credits, not invoices; one early approval allowed one GPU probe and two probe launches ran before the owner amended it.

Scope of the claims. The comparison covers two tasks, privileged state for Level 1, 180 training demonstrations per task from one operator channel, one training budget and one platform. Within that scope the evidence supports an advantage of BC-RNN over BC-MLP on Lift under both interval methods and on Can under the preregistered bootstrap only, gives no evidence that a diffusion head adds to a deterministic head with the same horizons, and shows that our point-cloud policy generalizes worse than our RGB policy to unseen cube positions. It supports no claim about real robots, other tasks, larger datasets or DP3 itself.

10 Conclusion

Under one preregistered protocol, with matched inputs, parameter counts, budgets and tuning, a diffusion policy over action chunks did not outperform a deterministic MLP that sees the same observation window and predicts the same chunks, on either Lift or Can. Its advantage over a one-step MLP on Can is not separated from that of an LSTM or of the chunked MLP, so the data give no evidence for the generative model beyond temporal context on these tasks, and even the separation from the one-step MLP depends on how the uncertainty over three seeds is computed. Diffusion was the most robust family to a large joint offset on Lift; on Can its drop was not separated from the chunked MLP's, and part of its robustness may come from its sampler's bound on its actions. It cost more than an order of magnitude more training and inference time than the chunked MLP. With images, visual BC and visual diffusion both reach the Lift ceiling, and our DP3-inspired point-cloud policy, which uses world-frame coordinates and no augmentation, generalized clearly worse than the matched RGB policy to unseen cube positions.

The most useful next steps follow from the limitations: more training seeds, harder tasks that are not saturated, more than one rollout per start, a BC baseline with the same action bound as the diffusion sampler, and a point-cloud arm with augmentation or an object-centred frame. Each would need its own preregistration and freeze before it is run.

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A Reproducibility

Software and data. Simulation: robosuite 1.5.1 (commit `51cc017`), MuJoCo 3.2.7; learning: robomimic 0.5.0 (commit `d309eae`), PyTorch 2.15.0, NumPy 2.4.6, diffusers 0.11.1 (scheduler reference only); Python 3.11.18. All versions are pinned in the repository's `uv.lock`. Every reported rollout ran in the container image `sha256:1f685638f28c`. The demonstrations are the Hugging Face `robomimic/robomimic_datasets` files at revision 74fa018, with the SHA-256 prefixes of Section 3, released under the MIT license according to the dataset card; `make download` fetches and verifies them, and they are not redistributed. The RoboForge code is released under the license Apache-2.0; robosuite, robomimic and MuJoCo keep their own licenses.

Splits and initial states. The training and validation episode IDs are in `data/splits/lift_ph.json` and `data/splits/can_ph.json` (Table 13). Each evaluation suite is a manifest with per-state hashes under `data/eval_states/<task>/<suite>/`; the held-out manifests' content SHA-256 prefixes are `e02498651b9d5bb8` (Lift) and `5f407748f5f261ef` (Can), as recorded in the Level 1 freeze. The robustness sets are derived deterministically from the base starts by `configs/eval/robustness.yaml`.

Frozen results. The Level 1 held-out analysis ran at commit `bf0a0fa`, the robustness analysis at `565c6e4`, the RGB analysis at `bc054bc` and the Level 2 analysis at `8c728f4`. Each study's freeze document, committed before its first governed run, lists the SHA-256 of every code and configuration file it depends on; a repository check compares those hashes with the working tree. Table 13 lists every frozen file this report reads, with its SHA-256 prefix. The evaluation summaries that a results file cites are read only after their hash matches the one recorded in that file.

Regenerating this report. Every table, figure and number in the text is generated from those files by `scripts/paper_assets.py`; the prose uses only generated macros. The generator never reads the interactive viewer's summaries or the training-run directories.

- `make paper-assets` regenerates `paper/generated/`, including an index with the SHA-256 of every output and of every source it read.
- `make paper-check` confirms that the committed generated files equal a fresh build byte for byte and lints the sections for typed numbers, placeholders and undefined macros.
- `make paper-bibcheck` resolves every DOI and arXiv identifier of the bibliography and compares titles and first authors (it needs network access).
- `make paper` builds `paper/main.pdf` with Tectonic from its cached bundle, without network access, and with a fixed source date so that the PDF is byte-identical across builds.

Commands and budgets. Training runs through `make train` and the suite orchestrator `make suite` with the committed plans (`configs/experiments/level1_core_lift.yaml` and `level1_core_can.yaml`): every run 100,000 steps, a checkpoint every 5,000 steps, every checkpoint on the 50 validation starts. Every rollout is capped at the 400-step horizon. The held-out tests and their analyses (`make eval-final`, `make analyze-final`, `make rgb-heldout`, `make pc-placement`, `make pc-heldout`, `make level2-analyze`) each ran once, after the freeze that governs it was committed, and must never be rerun. Each paid GPU launch was limited to one hour and ran under a committed approval with a dollar cap (Table 6).

What is not redistributed. Trained checkpoints, rollout records and per-episode arrays live in the training-run directories, which are not part of the repository. The committed evaluation summaries, selections and analyses suffice to regenerate every number here, but not to re-run a policy without retraining it. Retraining follows the committed configurations; Level 1 runs train on a CPU, and the RGB and Level 2 runs need one NVIDIA L4 GPU.

B Additional tables

Table 13: Frozen sources read by the report generator, with their SHA-256 prefixes.

Source	SHA-256 (prefix)	Content
`reports/session7/results/results.json`	`b8b097c38765c588`	Level 1 held-out analysis (frozen, run once)
`reports/session7/results/taxonomy.json`	`7a71518171c8303c`	Level 1 held-out failure taxonomy per episode
`reports/session7/robustness/analysis.json`	`8cca66a8a8a8cecf`	Level 1 robustness and ablation analysis (validation regime)
`reports/session8/rgb_results/results.json`	`57663b81aafc70e9`	RGB study held-out analysis (frozen, run once)
`reports/session9/level2_results/results.json`	`1d33c975e49df907`	Level 2 analysis, placement (H5) and held-out (frozen, run once)
`reports/session9/level2_results/ERRATA.md`	`280974a994408a6f`	Level 2 erratum (KNOWN_ISSUES KI-049), cited beside the row it corrects
`logs/session9/pc_train/launch_records/train_gpu_pc_20261109T130430.result.json`	`0ea097359ba49b37`	Level 2 seed 0 completing launch record (KI-049 correction)
`reports/compute/cost_ledger.jsonl`	`a4f92b9f1f4ae370`	Machine-readable cost ledger (launch, result and billing events)
`docs/FINAL_TEST_FREEZE.md`	`6a3c6874c4bc1857`	Level 1 held-out freeze (machine-readable block)
`docs/RGB_EXPERIMENT_FREEZE.md`	`47e7cb8ee70743db`	RGB study freeze
`docs/LEVEL2_EXPERIMENT_FREEZE.md`	`ffe5a03e98714465`	Level 2 freeze
`data/manifests/robomimic_v1.5_ph.json`	`34987db82f9a60a0`	Dataset manifest (robomimic v1.5 PH, Hugging Face revision, SHA-256)
`data/splits/lift_ph.json`	`e36d89afb6d0704a`	Lift demonstration split
`data/splits/can_ph.json`	`754967ce665fca00`	Can demonstration split
`reports/data_audit.json`	`1812ad718b506247`	Data audit (episode lengths, control period, action statistics)
`reports/session4/param_counts.json`	`0ec7b26a0a329737`	Exact parameter counts of the Level 1 families
`configs/eval/robustness.yaml`	`cfa933cac8719e32`	Preregistered robustness sets (RESEARCH_PROTOCOL §14)
`configs/eval/platform.yaml`	`a152eb41157f4b4d`	Canonical evaluation platform
`configs/experiments/level1_core_lift.yaml`	`d3de6cbcb14bcda1`	Level 1 Lift suite plan (budget, tuning trials)
`configs/experiments/level1_core_can.yaml`	`881b4fb7fd72a338`	Level 1 Can suite plan (budget, tuning trials)
`configs/experiments/level2_pointcloud.yaml`	`d91a546e1774779b`	Level 2 plan
`configs/models/diffusion_unet.yaml`	`ca181c22a4af5499`	Level 1 diffusion configuration (base; trials override the optimizer only)
`configs/models/pointcloud/diffusion_dp3_lift.yaml`	`b56f515dfd861568`	Level 2 point-cloud arm configuration
`reports/session6/suite/level1_core_lift.json`	`45c28837c1fd5079`	Level 1 Lift suite summary (validation selections, development numbers)
`reports/session7/suite/level1_core_can.json`	`213bb2c00de1a6f6`	Level 1 Can suite summary (validation selections, development numbers)
`uv.lock`	`327eaa2ea05a4e8f`	Locked Python environment (package versions)
`reports/session10/observatory/verification/lift-diffusion_unet-s0-lift_final_test_0000_t021_frontview_rgb.png`	`b6a165f72de1fb92`	Lift, diffusion seed 0, held-out start 0, frame 21: MuJoCo frontview render of the recorded state (Session 10 viewer-verification reference)
`reports/session10/observatory/verification/can-diffusion_unet-s0-can_final_test_0000_t060_frontview_rgb.png`	`149b860b17a2eca4`	Can, diffusion seed 0, held-out start 0, frame 60: MuJoCo frontview render of the recorded state (Session 10 viewer-verification reference)
`reports/session9/pc_feasibility/linux-aarch64-container-1f685638f28c/lift_overlay_demo_0_t029.png`	`fe163485aaccd6a2`	Lift demo 0, t = 29: the 512 sampled cloud points projected on the frontview and agentview frames (Session 9 feasibility, canonical container)
`reports/session7/heldout/` (26 files)	per file	Level 1 held-out evaluation summaries, each read only after its SHA-256 equals the one recorded in the Level 1 results file (D-096)

Table 14: Level 1 held-out results per selected checkpoint: training step, successes, Wilson 95 % interval and mean first-success step.

Family	Seed	Step	k/n	Wilson 95 %	First success	Errors
Lift						
BC-MLP	0	15,000	44/50	[76.2, 94.4]	54.2	0
BC-MLP	1	5,000	46/50	[81.2, 96.8]	48.7	0
BC-MLP	2	5,000	46/50	[81.2, 96.8]	47.3	0
BC-RNN	0	15,000	50/50	[92.9, 100.0]	43.5	0
BC-RNN	1	10,000	50/50	[92.9, 100.0]	42.3	0
BC-RNN	2	5,000	50/50	[92.9, 100.0]	42.5	0
BC-MLP-chunk	0	15,000	49/50	[89.5, 99.6]	45.0	0
BC-MLP-chunk	1	5,000	50/50	[92.9, 100.0]	43.3	0
BC-MLP-chunk	2	10,000	48/50	[86.5, 98.9]	43.3	0
Diffusion U-Net	0	100,000	49/50	[89.5, 99.6]	44.9	0
Diffusion U-Net	1	95,000	48/50	[86.5, 98.9]	44.1	0
Diffusion U-Net	2	80,000	49/50	[89.5, 99.6]	43.9	0
Random (uniform)	–	–	0/50	[0.0, 7.1]	–	0
Can						
BC-MLP	0	55,000	34/50	[54.2, 79.2]	121.4	0
BC-MLP	1	15,000	44/50	[76.2, 94.4]	116.3	0
BC-MLP	2	55,000	37/50	[60.4, 84.1]	117.4	0
BC-RNN	0	30,000	46/50	[81.2, 96.8]	114.2	0
BC-RNN	1	25,000	50/50	[92.9, 100.0]	118.3	0
BC-RNN	2	35,000	48/50	[86.5, 98.9]	110.6	0
BC-MLP-chunk	0	95,000	48/50	[86.5, 98.9]	104.6	0
BC-MLP-chunk	1	85,000	46/50	[81.2, 96.8]	106.2	0
BC-MLP-chunk	2	30,000	50/50	[92.9, 100.0]	109.2	0
Diffusion U-Net	0	65,000	46/50	[81.2, 96.8]	115.3	0
Diffusion U-Net	1	100,000	49/50	[89.5, 99.6]	114.3	0
Diffusion U-Net	2	90,000	48/50	[86.5, 98.9]	114.2	0
Random (uniform)	–	–	0/50	[0.0, 7.1]	–	0

Table 15: Validation success of the selected checkpoints beside their success on the fresh robustness base starts and on the held-out starts (%). Validation chose the checkpoints, so it is biased upward.

Family	Lift validation	Lift fresh base	Lift held-out	Can validation	Can fresh base	Can held-out
BC-MLP	97.3	98.0	90.7	82.7	69.3	76.7
BC-RNN	100.0	100.0	100.0	99.3	96.7	96.0
BC-MLP-chunk	99.3	100.0	98.0	100.0	95.3	96.0
Diffusion U-Net	100.0	99.3	97.3	98.7	92.0	95.3

Table 16: H4: per-family drops and between-family differences on the perturbation sets, with the bootstrap 95 % interval.

Contrast	Estimate (pp) [95 % CI]	Outcome
Lift		
drop of BC-MLP, base − cube shift	+45.3 [+30.7, +60.7]	separated (+)
drop of BC-RNN, base − cube shift	+4.0 [+0.0, +12.0]	not separated
drop of BC-MLP-chunk, base − cube shift	+45.3 [+24.7, +66.7]	separated (+)
drop of Diffusion U-Net, base − cube shift	+14.0 [+6.0, +22.7]	separated (+)
drop of BC-MLP, base − arm offset	+75.3 [+64.7, +85.3]	separated (+)
drop of BC-RNN, base − arm offset	+50.0 [+36.0, +64.0]	separated (+)
drop of BC-MLP-chunk, base − arm offset	+51.3 [+38.7, +64.0]	separated (+)
drop of Diffusion U-Net, base − arm offset	+11.3 [+2.7, +20.7]	separated (+)
Diffusion U-Net − BC-MLP on cube shift	+32.7 [+14.7, +50.7]	separated (+)
Diffusion U-Net − BC-RNN on cube shift	−10.7 [−21.3, +0.0]	not separated
Diffusion U-Net − BC-MLP-chunk on cube shift	+30.7 [+9.3, +52.7]	separated (+)
BC-RNN − BC-MLP on cube shift	+43.3 [+26.0, +62.0]	separated (+)
Diffusion U-Net − BC-MLP on arm offset	+65.3 [+52.0, +78.0]	separated (+)
Diffusion U-Net − BC-RNN on arm offset	+38.0 [+24.0, +52.0]	separated (+)
Diffusion U-Net − BC-MLP-chunk on arm offset	+39.3 [+25.3, +53.3]	separated (+)
BC-RNN − BC-MLP on arm offset	+27.3 [+13.3, +42.7]	separated (+)
drop Diffusion U-Net − drop BC-MLP on cube shift	−31.3 [−48.0, −15.3]	separated (−)
drop Diffusion U-Net − drop BC-RNN on cube shift	+10.0 [−0.7, +20.7]	not separated
drop Diffusion U-Net − drop BC-MLP-chunk on cube shift	−31.3 [−54.0, −10.0]	separated (−)
drop BC-RNN − drop BC-MLP on cube shift	−41.3 [−58.0, −25.3]	separated (−)
drop Diffusion U-Net − drop BC-MLP on arm offset	−64.0 [−76.7, −50.7]	separated (−)
drop Diffusion U-Net − drop BC-RNN on arm offset	−38.7 [−53.3, −24.0]	separated (−)
drop Diffusion U-Net − drop BC-MLP-chunk on arm offset	−40.0 [−54.0, −26.0]	separated (−)
drop BC-RNN − drop BC-MLP on arm offset	−25.3 [−40.0, −11.3]	separated (−)
Can		
drop of BC-MLP, base − arm offset	+39.3 [+26.0, +52.0]	separated (+)
drop of BC-RNN, base − arm offset	+43.3 [+30.0, +56.7]	separated (+)
drop of BC-MLP-chunk, base − arm offset	+16.0 [+6.0, +28.0]	separated (+)
drop of Diffusion U-Net, base − arm offset	+20.7 [+8.0, +34.0]	separated (+)
Diffusion U-Net − BC-MLP on arm offset	+41.3 [+26.0, +57.3]	separated (+)
Diffusion U-Net − BC-RNN on arm offset	+18.0 [+3.3, +32.7]	separated (+)
Diffusion U-Net − BC-MLP-chunk on arm offset	−8.0 [−23.3, +8.7]	not separated
BC-RNN − BC-MLP on arm offset	+23.3 [+8.7, +38.7]	separated (+)
drop Diffusion U-Net − drop BC-MLP on arm offset	−18.7 [−36.0, −0.7]	separated (−)
drop Diffusion U-Net − drop BC-RNN on arm offset	−22.7 [−39.3, −5.3]	separated (−)
drop Diffusion U-Net − drop BC-MLP-chunk on arm offset	+4.7 [−10.7, +20.7]	not separated
drop BC-RNN − drop BC-MLP on arm offset	+4.0 [−12.7, +21.3]	not separated

Table 17: Sensitivity analysis, not preregistered (D-095): every declared difference of the held-out studies with its frozen bootstrap interval and a seed-level t interval (degrees of freedom in brackets). $*$: the seed-level verdict differs from the frozen one. The frozen verdict is the reported result.

Comparison	Frozen bootstrap (pp)	Seed-level t (pp, df)	
Level 1 held-out, Lift			
H1: BC-MLP − Random (uniform)	+90.7 [+83.3, +96.7], separated (+)	[+84.9, +96.4] (2.0), separated (+)	
H1: BC-RNN − Random (uniform)	+100.0 [+100.0, +100.0], separated (+)	undefined (no seed variance)	
H1: BC-MLP-chunk − Random (uniform)	+98.0 [+94.0, +100.0], separated (+)	[+93.0, +103.0] (2.0), separated (+)	
H1: Diffusion − Random (uniform)	+97.3 [+93.3, +100.0], separated (+)	[+94.5, +100.2] (2.0), separated (+)	
H2: Diffusion − BC-MLP	+6.7 [−0.7, +14.7], not separated	[+1.9, +11.5] (2.9), separated (+)	$*$
H2: Diffusion − BC-RNN	−2.7 [−6.7, +0.0], not separated	[−5.5, +0.2] (2.0), not separated	
H2-c: Diffusion − BC-MLP-chunk	−0.7 [−5.3, +4.7], not separated	[−4.8, +3.4] (3.2), not separated	
H3: BC-RNN − BC-MLP	+9.3 [+3.3, +16.7], separated (+)	[+3.6, +15.1] (2.0), separated (+)	
Level 1 held-out, Can			
H1: BC-MLP − Random (uniform)	+76.7 [+62.7, +90.0], separated (+)	[+51.2, +102.2] (2.0), separated (+)	
H1: BC-RNN − Random (uniform)	+96.0 [+89.3, +100.0], separated (+)	[+86.1, +105.9] (2.0), separated (+)	
H1: BC-MLP-chunk − Random (uniform)	+96.0 [+88.7, +100.0], separated (+)	[+86.1, +105.9] (2.0), separated (+)	
H1: Diffusion − Random (uniform)	+95.3 [+89.3, +100.0], separated (+)	[+87.7, +102.9] (2.0), separated (+)	
H2: Diffusion − BC-MLP	+18.7 [+4.7, +32.7], separated (+)	[−4.5, +41.8] (2.4), not separated	$*$
H2: Diffusion − BC-RNN	−0.7 [−8.0, +6.7], not separated	[−9.0, +7.6] (3.7), not separated	
H2-c: Diffusion − BC-MLP-chunk	−0.7 [−8.7, +7.3], not separated	[−9.0, +7.6] (3.7), not separated	
H3: BC-RNN − BC-MLP	+19.3 [+5.3, +34.0], separated (+)	[−2.8, +41.5] (2.6), not separated	$*$
RGB study held-out, Lift			
Visual diffusion − Visual BC-chunk	+0.7 [+0.0, +4.0], not separated	[−2.2, +3.5] (2.0), not separated	
Level 2 placement, Lift			
H5: point cloud − RGB on cube shift	−41.3 [−54.0, −28.0], separated (−)	[−57.3, −25.4] (2.3), separated (−)	
drop of Point-cloud diffusion (base − cube shift)	+43.3 [+30.0, +56.7], separated (+)	[+25.9, +60.8] (2.0), separated (+)	
drop of Visual diffusion (base − cube shift)	+2.0 [+0.0, +6.7], not separated	[−3.0, +7.0] (2.0), not separated	
point cloud − RGB on base	+0.0 [+0.0, +0.0], not separated	undefined (no seed variance)	
drop point cloud − drop RGB	+41.3 [+28.0, +54.0], separated (+)	[+25.4, +57.3] (2.3), separated (+)	
Level 2 held-out, Lift			
point cloud − RGB	+0.0 [+0.0, +0.0], not separated	undefined (no seed variance)	

Table 18: Sensitivity analysis, not preregistered (D-095), for H4 and the ablations; as in Table 17.

Comparison	Frozen bootstrap (pp)	Seed-level t (pp, df)	
H4, Lift			
drop of BC-MLP, cube shift	+45.3 [+30.7, +60.7], separated (+)	[+18.0, +72.7] (2.0), separated (+)	
drop of BC-RNN, cube shift	+4.0 [+0.0, +12.0], not separated	[−9.1, +17.1] (2.0), not separated	
drop of BC-MLP-chunk, cube shift	+45.3 [+24.7, +66.7], separated (+)	[−4.4, +95.1] (2.0), not separated	*
drop of Diffusion, cube shift	+14.0 [+6.0, +22.7], separated (+)	undefined (no seed variance)	
drop of BC-MLP, arm offset	+75.3 [+64.7, +85.3], separated (+)	[+72.5, +78.2] (2.0), separated (+)	
drop of BC-RNN, arm offset	+50.0 [+36.0, +64.0], separated (+)	[+32.8, +67.2] (2.0), separated (+)	
drop of BC-MLP-chunk, arm offset	+51.3 [+38.7, +64.0], separated (+)	[+37.0, +65.7] (2.0), separated (+)	
drop of Diffusion, arm offset	+11.3 [+2.7, +20.7], separated (+)	[−0.1, +22.8] (2.0), not separated	*
Diffusion − BC-MLP on cube shift	+32.7 [+14.7, +50.7], separated (+)	[−3.0, +68.3] (2.0), not separated	*
Diffusion − BC-RNN on cube shift	−10.7 [−21.3, +0.0], not separated	[−23.1, +1.7] (2.2), not separated	
Diffusion − BC-MLP-chunk on cube shift	+30.7 [+9.3, +52.7], separated (+)	[−18.9, +80.2] (2.0), not separated	*
BC-RNN − BC-MLP on cube shift	+43.3 [+26.0, +62.0], separated (+)	[+11.8, +74.9] (2.5), separated (+)	
Diffusion − BC-MLP on arm offset	+65.3 [+52.0, +78.0], separated (+)	[+56.5, +74.1] (3.9), separated (+)	
Diffusion − BC-RNN on arm offset	+38.0 [+24.0, +52.0], separated (+)	[+23.6, +52.4] (2.9), separated (+)	
Diffusion − BC-MLP-chunk on arm offset	+39.3 [+25.3, +53.3], separated (+)	[+27.5, +51.1] (3.3), separated (+)	
BC-RNN − BC-MLP on arm offset	+27.3 [+13.3, +42.7], separated (+)	[+13.2, +41.5] (3.3), separated (+)	
drop Diffusion − drop BC-MLP, cube shift	−31.3 [−48.0, −15.3], separated (−)	[−58.7, −4.0] (2.0), separated (−)	
drop Diffusion − drop BC-RNN, cube shift	+10.0 [−0.7, +20.7], not separated	[−3.1, +23.1] (2.0), not separated	
drop Diffusion − drop BC-MLP-chunk, cube shift	−31.3 [−54.0, −10.0], separated (−)	[−81.1, +18.4] (2.0), not separated	*
drop BC-RNN − drop BC-MLP, cube shift	−41.3 [−58.0, −25.3], separated (−)	[−64.3, −18.3] (2.9), separated (−)	
drop Diffusion − drop BC-MLP, arm offset	−64.0 [−76.7, −50.7], separated (−)	[−74.7, −53.3] (2.2), separated (−)	
drop Diffusion − drop BC-RNN, arm offset	−38.7 [−53.3, −24.0], separated (−)	[−52.8, −24.5] (3.5), separated (−)	
drop Diffusion − drop BC-MLP-chunk, arm offset	−40.0 [−54.0, −26.0], separated (−)	[−52.1, −27.9] (3.8), separated (−)	
drop BC-RNN − drop BC-MLP, arm offset	−25.3 [−40.0, −11.3], separated (−)	[−41.9, −8.7] (2.1), separated (−)	
A1, Lift			
primary − trained-weights	+0.7 [+0.0, +2.7], not separated	[−2.2, +3.5] (2.0), not separated	
A2, Lift			
primary − ddpm100	−0.7 [−4.0, +0.0], not separated	[−3.5, +2.2] (2.0), not separated	
H4, Can			
drop of BC-MLP, arm offset	+39.3 [+26.0, +52.0], separated (+)	[+27.9, +50.8] (2.0), separated (+)	
drop of BC-RNN, arm offset	+43.3 [+30.0, +56.7], separated (+)	[+33.0, +53.7] (2.0), separated (+)	
drop of BC-MLP-chunk, arm offset	+16.0 [+6.0, +28.0], separated (+)	[−5.7, +37.7] (2.0), not separated	*
drop of Diffusion, arm offset	+20.7 [+8.0, +34.0], separated (+)	[+1.9, +39.5] (2.0), separated (+)	
Diffusion − BC-MLP on arm offset	+41.3 [+26.0, +57.3], separated (+)	[+25.8, +56.9] (3.6), separated (+)	
Diffusion − BC-RNN on arm offset	+18.0 [+3.3, +32.7], separated (+)	[+1.7, +34.3] (2.6), separated (+)	
Diffusion − BC-MLP-chunk on arm offset	−8.0 [−23.3, +8.7], not separated	[−29.2, +13.2] (3.7), not separated	
BC-RNN − BC-MLP on arm offset	+23.3 [+8.7, +38.7], separated (+)	[+12.5, +34.2] (3.2), separated (+)	
drop Diffusion − drop BC-MLP, arm offset	−18.7 [−36.0, −0.7], separated (−)	[−34.1, −3.2] (3.3), separated (−)	
drop Diffusion − drop BC-RNN, arm offset	−22.7 [−39.3, −5.3], separated (−)	[−38.2, −7.1] (3.1), separated (−)	
drop Diffusion − drop BC-MLP-chunk, arm offset	+4.7 [−10.7, +20.7], not separated	[−14.0, +23.3] (3.9), not separated	
drop BC-RNN − drop BC-MLP, arm offset	+4.0 [−12.7, +21.3], not separated	[−6.0, +14.0] (4.0), not separated	
A1, Can			
primary − trained-weights	−2.7 [−7.3, +0.7], not separated	[−5.5, +0.2] (2.0), not separated	
A2, Can			
primary − ddpm100	−3.3 [−10.0, +3.3], not separated	[−10.9, +4.3] (2.0), not separated	
A3, Can			
T_o=2 − T_o=1	−2.0 [−8.7, +4.0], not separated	[−9.1, +5.1] (3.2), not separated	